

6DATA007W DATA SCIENCE AND ANALYTICS
FINAL PROJECT
2025-26

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF DESIGN, CREATIVE AND DIGITAL INDUSTRIES (DCDI)

STAFF TEAM CONTACT DETAILS

MODULE LEADER:

- Dr Salma Chahed, chaheds@westminster.ac.uk

LECTURERS:

- Dr Salma Chahed, chaheds@westminster.ac.uk
- Ka-Ming Pang, ask-aeld@westminster.ac.uk
- Dr Joseph Nicholson, ask-aeld@westminster.ac.uk

PROJECT SUPERVISORS

- Most Computer Science & Engineering academic staff. Link to the University of Westminster directory for all staff details:

<https://www.westminster.ac.uk/about-us/our-people/academic-directory>

TABLE OF CONTENTS

1. Introduction.....	4
2. Aims & Learning Outcomes	5
2.1 Module Aims.....	5
2.2 Learning Outcomes (LOs).....	5
3. Project Topic.....	5
4. Working With a Supervisor	6
5. Module Delivery	7
5.1 Lectures and Workshops.....	7
5.2 Demonstrations	8
5.3 Study code of practice	8
6. Module Timeline	8
7. Assessments & Milestones	9
7.1 Summative assessments	9
7.2 Milestones	9
7.3 Assessment Completion, Submission & Marking	10
7.3.1 <i>Project Proposal and Specification (PPS) – Summative assessment.....</i>	<i>10</i>
7.3.2 <i>Interim Progress Demonstration (IPD)- Summative assessment.....</i>	<i>10</i>
7.3.3 <i>Final Year Project (FYP) - Summative assessment.....</i>	<i>11</i>
7.3.4 <i>Data collection and pre-processing / Code implementation demo - milestone.....</i>	<i>12</i>
7.3.5 <i>Initial project outputs - milestone.....</i>	<i>12</i>
7.4 Plagiarism.....	12
8. Other Submissions	13
8.1 Request for Project Supervision form	13
8.2 Ethics Forms.....	13
9. Bibliography.....	13

1. INTRODUCTION

The Final Project (FP) module is a significant part of the Computer Science and Software Engineering BSc degrees offered by the School of Computer Science & Engineering. It requires the student to combine previously acquired knowledge and techniques gained during their studies, with new skills and researched ideas to produce an extended piece of work, the Final Project.

The project topic is based on the student's own interests or can be chosen from a list of project areas/titles provided by the academic members of staff and available on the module's Blackboard. The topic should integrate the different strands related to the degree of study and apply them successfully to solve a real-world problem or tackle a research question. The project concept is not expected to be a wholly original idea, but it has to show resourcefulness, motivation, exercising initiative and ability for independent study. The project thus provides the opportunity to students to demonstrate their skills in exploring and understanding a topic in depth. It gives the chance to exercise skills, which are difficult to measure in other ways.

Students work independently under the guidance of a supervisor and are expected to develop their work following a systematic approach and applying scientific methodologies and techniques. They are expected to set up regular meetings together with their supervisor to review progress and discuss various aspects of the project. Student engagement during the year is taken into account in the module assessment.

The major expected output of the module is a data-driven solution or analytical models (most often), alternatively a research result. Either way, the project output is accompanied by a written report and is presented during an oral examination (a viva) at the end of the module.

The module learning method are based on a series of live lectures and workshops that are scheduled mainly in semester 1, every 1 or 2 weeks, taking place either online or onsite. They are designed to support the project development, and their scheduling is based on the project development lifecycle.

Important information, announcements and learning resources are disseminated through the Blackboard module site. Students are advised to engage with the module's Blackboard site as soon as possible. They are expected to attend the organised workshops, meet regularly with supervisors, keep track of the discussion board, post questions, share experience and pick up guidance given to others that might also be relevant to their work.

2. AIMS & LEARNING OUTCOMES

2.1 Module Aims

The aim of the FP module is to provide the student with an understanding of the skills, methodology and processes and enable them to implement a project starting from understanding the problem through to successful model development, documentation, and communication of findings. On completion of the module, the student will be equipped to work autonomously on real-world challenges.

2.2 Learning Outcomes (LOs)

By the end of the module the successful student will be able to:

- LO1 Select, justify and apply appropriate modelling and visualisation methods, techniques and tools for tackling a large problem
- LO2 Develop a project plan, organise own activities and manage time and workload within the given timescale
- LO3 Research and collate relevant information from various sources, critically evaluate the findings and formulate an industry-related problem
- LO4 Work autonomously to learn new skills and/or enhance existing knowledge and produce deliverables agreed with the supervisor
- LO5 Identify and take into account any legal, social, ethical EDI or professional issues related to the problem and its solution including sustainability considerations
- LO6 Assess a complex problem logically, develop a comprehensive solution taking into consideration the environmental and societal impact of their solutions and interpret related results in the original context of the problem
- LO7 Produce a coherent report that documents and critically evaluates the practical work, the new skills and the effectiveness of the project plan
- LO8 Demonstrate and defend their work at a viva voce examination

The module contributes to the following individual course LOs.

BSc Data Science and Analytics: L6.1, L6.2, L6.3, L6.6, L6.7, L6.8

3. PROJECT TOPIC

The project topic should be best matching the students' knowledge and expertise, as well as satisfying the module and course requirements. It should reflect the student's personal

interests, provide the opportunity to practice and expand existing skills and develop new ones. It must be chosen during the starting weeks of semester 1 if not before. The topic will be discussed and approved by the assigned Project Supervisor. It is advised to liaise with the supervisor and the teaching team to ensure that the necessary resources required for the successful completion of the project are available / can be made available.

It is expected that the project will have a significant practical element. Usually this comprises of the development of analytical models, or a data-driven solution. Alternatively, it may comprise of a practical study designed to solve a research problem. The latter type of projects is more rarely chosen and are generally more challenging.

Several Final Project topics are proposed by academic members of staff. A list can be found on the Blackboard site of the module. You may consider selecting one of the suggested projects, but this is not compulsory. Most students come up with their own project ideas. If you choose one of the proposed projects, make sure that you discuss your selection with the respective staff members as soon as possible.

4. WORKING WITH A SUPERVISOR

Every student is assigned a project supervisor in the start of the academic year. The allocation is based on staff availability. Students are asked to indicate supervisors of their choice by filling up the Supervision Request Form during Learning Weeks (LW) 1 and 2; the form is available on Blackboard in LW1. The process however does not guarantee that one of the requested supervisors will be assigned to the student, but it facilitates the supervisors' allocation. Students that have a supervision agreement with a specific staff member will be prioritised during the supervision allocation, provided that they will present evidence of their agreement (copy of email) in the Supervision Request. Allocation of supervisors is completed and communicated at the end of Learning Week 3.

The supervisor has a crucial role in the development of the final year project. Students are advised to maintain a regular contact with their supervisor. It is the student's responsibility to keep in touch with their supervisor and update them on their progress.

The duties of the supervisor include: help students decide on the area of their project; guide students on and take responsibility for the ethical aspects of their project; guide students on how to follow the project plan work during the year; check the students' progress throughout the year; provide informed discussion and guidance about the project.

Students are encouraged to contact project supervisor, using email. Supervision meetings are conducted on-line or in person. They should be regular (approx. every 1-2 weeks) throughout the entire academic year. In some cases, the project supervisor, or the module leader will request meetings that relate to formative assessments that students are expected to attend.

Students are advised to record a log for every meeting with their supervisor with main points raised and agreed. A copy of this log should be available to both the student and the supervisor.

The Module Leader in communication with the Head of School carries out the allocation of the supervisor. It is based on the project topic, the student's previous engagement with staff members and their preference (if submitted), staff expertise and availability. The supervisor is not expected to be an expert on the technical aspects of the project area but should be considered as the project manager and mentor who keeps track and advises on the project feasibility, progress and lifecycle. Members of staff other than the supervisor are available to give specific technical help and advice, if required.

5. MODULE DELIVERY

5.1 Lectures and Workshops

The FP, unlike any of the other modules, provides students the opportunity to work independently. A programme with timetabled sessions relating to all aspects of the project lifecycle will be delivered live, on-line or in campus during the academic year to help students build their projects. Most lectures and workshops will take place on Ms Teams (accessible via Blackboard). They are scheduled every 1-2 weeks (listed in the timetable) on Fridays, morning (11:00 -13:00), unless notified otherwise.

Module information and learning materials are disseminated via Blackboard, module announcements and module emails. It is important to keep track of all relevant communication, check Blackboard announcements and contents.

The topics covered in lectures and workshops are:

1. Introduction to the Final Project module
2. Project success toolkit: from ideation to management and sharing
3. Ethics, Sustainability and other considerations
4. Poster creation
5. Literature review and referencing system
6. Interim Progress Demonstration (IPD): Introduction and Q&A
7. Academic report writing and viva preparation
8. Final Year Project (FYP): Introduction and Q&A

5.2 Demonstrations

Two Demo sessions are timetabled but not graded. These represent good opportunities for students to collect constructive feedback from their peers and wider audience.

The first Demonstration session takes place in week 10 of semester 1. Students present their initial project idea and evidence of some background research. The posters enable them to visually convey information about their work.

The second Demonstration session takes place in week 5 of semester 2. Students present an overview of their project progress using appropriate tools and support.

5.3 Study code of practice

Students are expected to respect the following list of professional conduct:

- Read and act on Blackboard announcements and emails in a timely manner
- Ensure their SRS contact details are current.
- Write emails in a formal professional style, addressing the staff member who they email with their names and closing by signing with their names.
- Be in communication with their supervisor throughout the year
- Attend regular supervision meetings
- Request permission for recording meetings beforehand
- Keep supervisory logs to minutes the meetings and record verbal feedback
- Deliver/attend all formative and summative assessments.

6. MODULE TIMELINE

As a rough guide, the final year project module should take around 400 hours of your time. This time should be distributed reasonably giving full considerations to other modules on your timetable. Workshops, Lectures and Demonstrations are the only scheduled activities, otherwise students are responsible for scheduling the majority of their final year project work. They are advised to assign time every week throughout the year for working on the development of the project, starting from the first week of term. There are deliverables (formative and summative assessments and recommended milestones – see section 7), designed to help students build their project and obtain valuable and timely feedback from supervisors and, at times, from peers.

The module Timeline (available in Blackboard) indicates module activities other than the student's own study time and the scheduled workshops. It indicates milestones for the supervisors and the students. Students are advised to keep this timeline available and consult it frequently. It indicates: when documents, or assessments need to be submitted,

suggested weeks to meet the supervisors (note, students may seek meetings at other times depending on individual needs and progress), expected dates for obtaining feedback and marks, etc.

7. ASSESSMENTS & MILESTONES

The assessment covers both the product delivered by the student at the end of the module, and the process by which this product was developed. In addition, students are required to regularly demonstrate the extent of the development of their work.

7.1 Summative assessments

They are marked assessments that contribute to the module's mark. They are:

- Project Proposal and Specification (PPS) – 15% module mark (deadline 13/11/25)
A poster presentation that enables students to engage with a project idea; carry out background research and initial investigations.
(LO2, LO3).
- Interim Progress Demonstration (IPD) – 15% module mark (deadline 05/02/26)
A presentation, a progress report and evidence of initial output from practical work (e.g., data collection, data pre-processing, codes, initial model(s), software prototype) that evidence that students can progress smoothly toward the development of their Final Year Project deliverables.
(LO1, LO2, LO3, LO4, LO6).
- Final Year Project (FYP) – 70% module mark (deadline 23/04/26)
A report, the final output from practical work and viva (oral examination) that enable students to demonstrate the competence they gained in delivering a self-initiated project, communicate it effectively in written and oral forms and the competences required to achieve the degree of study.
(LO1-LO8).

7.2 Milestones

These are not marked components, but their timely completion along with the completion of the formative assessments demonstrates student engagement, proficiency and performance. They are designed as check points in the development of the project.

- Data collection and pre-processing / First code implementation demo (Semester 1, LW 10/11/12)

A first demo explaining what and how relevant data was collected and pre-processed, presenting research developed to date and/or capturing key functionality of the software.

- Initial project outputs (Semester 2, LW 1/2)

An outline of the finalised methodological approach and outputs of data analysis. A close-to-final list of project requirements if a software is being developed.

7.3 Assessment Completion, Submission & Marking

7.3.1 Project Proposal and Specification (PPS) – Summative assessment

The Project **poster** includes project problem, aims and objectives, initial background research on similar projects and applications, an outline of the proposed methodological approach, data presentation and/or list of requirements, a list of tools and software required to develop the project, a project timeline in the form of Gantt chart or other, potential risks and how to mitigate them, and references. The guidelines for the preparation of the poster are made available to students on Blackboard to facilitate its development.

The submission of the PPS is done via Blackboard. Submission links can be found under Assessment. A feedback form including a marking rubric, which is used to assess the PPS, is also made available.

Students are expected to attend a **20-min live poster presentation**, i.e. an oral examination of the project proposal using the poster as support. Two members of staff, normally the project supervisor and another member of staff, assess the student's performance. During the viva, the student is expected to demonstrate their project idea, initial background research and approach and how they intend to manage their project.

Two academics, including the project supervisor, mark the PPS. Both assessors first mark independently the poster and after the **presentation** agree to a mark.

7.3.2 Interim Progress Demonstration (IPD)- Summative assessment

The Interim Progress Demonstration comprises of:

i) a **progress report** that includes changes from the original PPS; work completed to date; key challenges; revised project timeline clarifying outstanding tasks and associated risks; a **link to a 20-minute video demo** that demonstrates any practical work progress (code, prototype, etc.).

ii) a **practical output** – evidence of data collection and/or generation, pre-processing and analysis using relevant tools and/or a prototype of the software/system being developed capturing key functionality. Project Code should be available via student's **GitHub account** (i.e., provide URL to your public repo).

Feedback on the report and the practical output should assist the student develop further the project requirements, if needed, and assess the practical work progress.

7.3.3 Final Year Project (FYP) - Summative assessment

A successful Final Project is a combination of practical work and a report detailing the project development and a critical analysis of the resulting research output/model(s)/software/application. The assessment comprises of three elements:

- i) the Final Project Report (FPR),
- ii) the Final Project Code - or Research Results- (FPC), and
- iii) the Final Project Viva (FPV).

The **FPR** details the work carried out to complete the project. It is assessed for logical structure, coherence and relevance. It is also assessed for clear documentation on the scope, depth and complexity of the work; the background research; the development of new and existing skills; the selection and application of appropriate methods, techniques and tools; the different aspects of the development and analysis process; any legal, social, ethical or professional issues relevant to the work; the reflection and critical evaluation of the work.

The **FPC** (or practical research output of the research*) is assessed in terms of level of complexity, performance (rigour*) and originality. For projects containing a software development component, the FPC is also assessed in terms of stability & robustness. Project Code should be available via student's **GitHub account**.

The FPR and the FPC are submitted via Blackboard using the links provided under Assessment. Students are expected to also submit a brief (5-10 minute) **video demo** of datasets, model(s) and results (and the software to introduce its main functionality and features if applicable). This is designed to assist the assessment. It is advisable that the demo is uploaded on YouTube or another relevant platform and that a link to it is provided in the FPR report.

The **FPV** is an oral examination of the project output. Students are expected to attend a 1-hour viva. Two members of staff, normally the project supervisor and another member of staff, assess the student's performance. During the viva, the student is expected to demonstrate their practical output (i.e., model(s) and related findings, software, working prototype, research outcomes) and to be tested on their understanding of relevant issues. Additionally, students are expected to demonstrate understanding of the wider context and propose further work.

Supervisor and moderator first mark independently the report and code and after the viva agree to a mark. If there is a disagreement between the marks of the supervisor and the moderator, a third examiner is asked to assess the FPR and FPC.

* for research projects

In case students fail to submit the FPR and FPC, or attend the FYV, they will have to follow the procedure for MC application. All three elements have to be attempted by the student for the FYP to be marked.

7.3.4 Data collection and pre-processing / Code implementation demo - milestone

A demonstration of what relevant datasets were collected and pre-processed, and how it was done. In case of software/application development, code implementation capturing key functionalities developed to date should also be demonstrated. The student is expected to request a supervision meeting for this milestone during the last weeks of Semester 1. This is not marked. The supervisor provides feedback orally.

7.3.5 Initial project outputs - milestone

The methodological approach should be well defined by this stage. Students should have a good understanding of the data and the outputs of data analysis. If a software/application is being developed, project requirements should be well defined by this stage, and grouped into functional/non-functional, also classified as essential, desired and luxury. The student is expected to request a supervision meeting to discuss the initial project outputs with the supervisor during the first two weeks of Semester 2. The outputs are not marked. The supervisor provides feedback orally during the meeting that should help the IPD development.

7.4 Plagiarism

It is very important to understand what plagiarism is and how serious its consequences could be for the project assessment. Students are referred to the [University Academic Regulations](#) (part 3) with regard to the issue of plagiarism. Check also the [Academic Matters, Plagiarism](#) webpage.

If you need to take some citation from someone else's work, remember to draw directly on the work of experts in their field of study, which should be related to your problem domain. Try to limit direct quotations to a few lines only and reference ALL instances in both the text and include sources in the reference list, including detailed addresses of Internet references and date the Internet page was accessed.

The University of Westminster recommends the [cite them right](#) Harvard style of referencing. All projects will be subject to Turnitin investigation for plagiarism.

8. OTHER SUBMISSIONS

8.1 Request for Project Supervision form

The Request for Project Supervision form is an online form that indicates the subject area the student has selected for their FYP and any agreements made with academic staff for their supervision. The form is submitted via a Blackboard link that is made available in the beginning of the academic year. It is used to guide and facilitate the allocation of supervisors. The allocation of supervisors is based on the student's form submission, expertise and availability of supervisors, etc. (see section 4).

8.2 Ethics Forms

The University recommends all researchers at all levels to be familiar with and follow the guidance for research ethics and research good practice of their relevant professional bodies. The University expects its researchers to conduct its research ethically and the University is concerned to protect the rights, dignity, health, safety and privacy of research participants, the welfare of animals and the integrity of the environment. The University is also concerned to protect the health, safety, rights and academic freedom of researchers and the reputation of the University as a centre for properly conducted, high quality research.

Each project therefore requires ethics approval. Note that the supervisor is responsible for the ethical issues relating to each supervised final year project. Students are expected to access the Ethics form via Blackboard (Link to Ethics form is available in document *6DATA007W PPS Guidelines.docx*) and liaise with the module tutors during the relevant workshops for completing the form. Once completed, the form should be signed by both student and supervisor, and submitted using the same submission link as the poster.

9. BIBLIOGRAPHY

The recommended Reading Lists can be found on Blackboard; it is updated regularly.

[6DATA007W Data Science and Analytics Final Project](#)

[Data Science and Business Intelligence background reading](#)